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16 August 1982
SUPERSEDING
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20 May 1974

MILITARY SPECIFICATION

ROLLER, TRANSOM, BAILEY BRIDGE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of transom roller for use in Bailey bridge erection.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-S-741	- Steel, Carbon: Structural Shapes; Plates; and Bars.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS24665	- Pin, Cotter (Split).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/ASTM A 108	- Steel Bars, Carbon, Cold-Finished, Standard Quality.
ANSI/ASTM A 120	- Pipe, Steel, Black and Hot-Dipped, Zinc-Coated (Galvanized), Welded and Seamless, for Ordinary Use.
ANSI/AWS D1.1	- Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The roller shall be as shown in figure 1 and as specified herein.

3.2 Material. Material shall be as specified herein (see 6.4 and 6.5).

3.2.1 Structural steel. Structural steel shall conform to QQ-S-741, 12 pounds.

3.2.2 Cotter pins. Cotter pins shall conform to MS24665-497. Each roller shall be provided with an extra cotter pin.

3.2.3 Steel bar. Steel for pins and bars shall conform to ANSI/ASTM A 108, UNS G10200, cold drawn and stress relieved.

3.2.4 Pipe. Pipe shall conform to ANSI/ASTM A 120, standard weight, except that ends shall not be threaded and hydrostatic test is not required.

3.2.5 Bronze. Bronze for bushings shall conform to QQ-C-390, alloy 905, type I, condition as cast.

3.3 Identification marking. The roller shall be identified in accordance with MIL-STD-130.

3.4 Treatment and painting. The portions of the roller normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type A. Color shall be in accordance with FED. STD. No. 595, color No. 36231.

3.5 Workmanship.

3.5.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.5.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the transon rollers to conform to the examination requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the transom rollers shall be at the option of the contractor.

3.5.3 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.
ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4 Inspection procedure.

4.4.1 Examination. The roller shall be examined as specified herein for the following defects:

Major

101. Roller not in accordance with figure 1.
102. Materials not as specified.
103. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.
202. Treatment and painting not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

104.	Materials and containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.
105.	Quantity packed together exceeds the weight or size limitation of the box for level A or B.
106.	Rollers not blocked and braced in accordance with the referenced document as specified for level A or B.
107.	Packing not in accordance with the referenced document for commercial.
108.	Marking missing, illegible, incorrect, or incomplete for level A, B or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. The rollers shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. The contents shall be blocked and braced within the box to prevent free movement or damage in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2) strapping shall be finish B. When specified (see 6.2) strapping shall be finish A.

5.1.2 Level B. The rollers shall be packed as specified in 5.1.1, except boxes shall be domestic type. As an alternate, a box conforming to PPP-B-636, V3c, V11c, V13, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitations of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.3 Commercial. The rollers shall be packed in accordance with MIL-STD-1188.

5.2 Marking (see 6.3).

5.2.1 Military. Shipping containers for level A or B shall be marked in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The transom roller is intended for use in the erection of portable panel Bailey-type bridges.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Welding process required when other than as specified (see 3.5.2).

c. Degree of packing required (see 5.1).

d. When other than finish B strapping is required (see 5.1.1).

6.3 Preservation/packing level marking. Preservation is considered unnecessary. For purposes of preservation/packing level marking, preservation shall be designated level A.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

6.5 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bids, requests for proposal, and contracts the following statement:

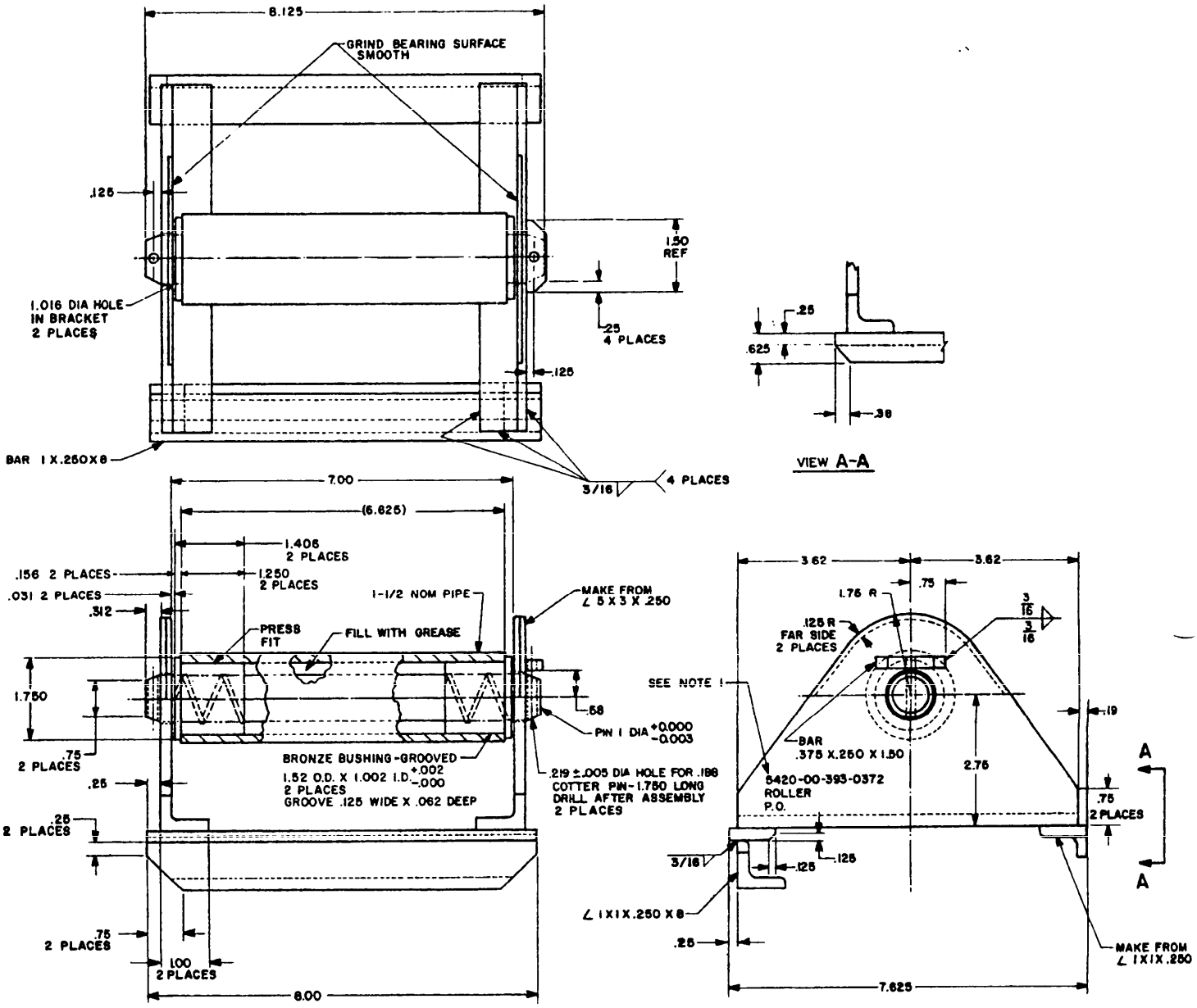
"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed." (see 3.2).

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NOTES:

1. STEEL STAMP, .250 HIGH CHARACTERS, AT APPROXIMATE LOCATION SHOWN.
2. ALL DIMENSIONS ARE IN INCHES.
3. TOLERANCES, UNLESS OTHERWISE SHOWN: .XX ± .06 .XXX ± .030

FIGURE 1. Transom roller.

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